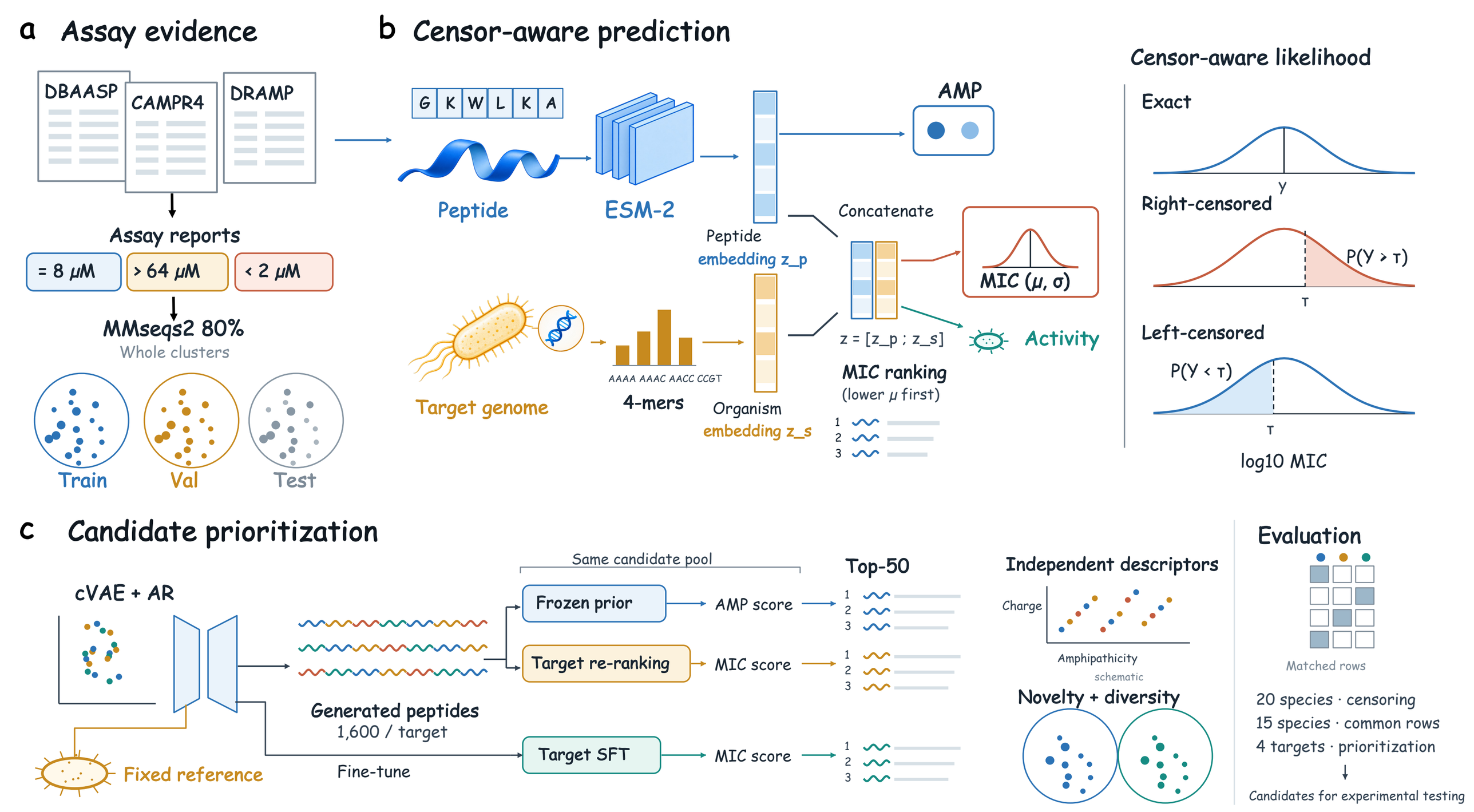

a Assay evidence
DBAASP
CAMPR4
DRAMP
Assay reports
= 8 μM
> 64 μM
< 2 μM
MMseqs2 80%
Whole clusters
Train
Val
Test
b Censor-aware prediction
G K W L K A
Peptide
ESM-2
AMP
Peptide embedding z_p
Concatenate
MIC (μ, σ)
Activity
z = [z_p ; z_s]
MIC ranking
(lower μ first)
1
2
3
Target genome
AAAA AAAC AACC CCGT
4-mers
Organism embedding z_s
Censor-aware likelihood
Exact
y
Right-censored
P(Y > τ)
τ
Left-censored
P(Y < τ)
τ
log10 MIC
c Candidate prioritization
cVAE + AR
Fixed reference
Generated peptides
1,600 / target
Fine-tune
Same candidate pool
Frozen prior
AMP score
Target re-ranking
MIC score
Target SFT
MIC score
Top-50
1
2
3
Independent descriptors
Charge
Amphipathicity
schematic
Novelty + diversity
Evaluation
Matched rows
20 species · censoring
15 species · common rows
4 targets · prioritization
Candidates for experimental testing